

RAKR of HDR Brachytherapy Source



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Contents

1	Objective	1
2	Apparatus	1
3	Theory	1
3.1	Introduction to Brachytherapy	1
3.2	Categorization of Brachytherapy	1
3.2.1	Classification by Dose Rate	1
3.2.2	Classification by Source Placement	2
3.2.3	Classification by Implant Duration	2
3.3	Source Specification Methods Currently in Use	2
3.3.1	Activity of the Radionuclide	2
3.3.2	Exposure Rate at a Reference Point in Free Space	2
3.3.3	Equivalent Mass and Apparent Activity	2
3.3.4	Air-Kerma Rate at a Reference Point in Free Space	3
3.4	Brachytherapy Radioactive Sources	3
3.4.1	Encapsulation and Design Integrity	3
3.4.2	Photonic Sources	3
3.4.3	Source Strength Metrics	3
3.5	Brachytherapy Equipment: Siemens Healthineers BRAVOS	3
3.6	Dosimetry via Well-Type Ionization Chamber	4
3.6.1	The Well Chamber Design	4
3.6.2	Measurement Formalism	4
3.7	HDR Afterloader Mechanics	4
4	Observation	4
4.1	Determination of Maximum Response Position (Sweet Spot)	5
4.2	Evaluation of Source Strength	5
4.2.1	Derivation of Correction Factors	6
4.2.2	RAKR and Source Activity Computation	6
4.3	Timer Error Determination	6
4.4	Timer Linearity and End Error Assessment	7
5	Conclusion	7

1 Objective

The primary objectives of this experiment are:

- To measure the Air Kerma Strength (AKS) or Reference Air Kerma Rate (RAKR) of the High-Dose Rate Brachytherapy source (Ir-192).
- Find the timer error and the percentage of error in the calculation timer error.
- Find the percentage of error in timer linearity and end error.

2 Apparatus

The following instruments and materials were utilized to conduct this experiment:

- High Dose Rate (HDR) Brachytherapy Delivery Unit
- Well-type Ionization Chamber Dosimetry System
- Precision Electrometer with required connecting triaxial cables
- Calibrated ambient monitoring devices (Thermometer and Barometer)

3 Theory

3.1 Introduction to Brachytherapy

Brachytherapy, deriving its name from the Greek word "brachys" meaning short distance, is an advanced modality of radiation oncology that involves the precise placement of encapsulated radioactive sources directly inside or immediately adjacent to a tumor volume [1]. The paramount clinical advantage of this modality lies in its adherence to the inverse square law; it delivers an extremely conformal, tumor-ablative radiation dose while ensuring a rapid dose fall-off that sharply spares surrounding critical healthy organs [2]. Depending on the clinical indication, brachytherapy can function as an exclusive, definitive treatment or as a highly targeted sequential boost complementing External Beam Radiotherapy (EBRT).

Before any brachytherapy source is released for clinical patient treatment, independent physics verification of the source strength is legally and clinically mandated. Historically categorized by equivalent mass of radium, modern dosimetry explicitly quantifies source strength as the Air Kerma Strength (S_K) or the Reference Air Kerma Rate (RAKR) [3]. Ascertaining the RAKR via absolute dosimetric measurement ensures that the actual radioactive output matches the manufacturer's certificate within a stringent tolerance (typically $\pm 3\%$), maintaining unbroken traceability to Primary Standard Dosimetry Laboratories (PSDLs) [4]. Furthermore, High Dose Rate (HDR) treatments deliver massive localized doses in fractions of a minute; consequently, the electromechanical accuracy of the remote afterloader's timer and transit mechanism is of profound dosimetric consequence.

3.2 Categorization of Brachytherapy

Brachytherapy interventions are broadly stratified by the rate of dose delivery, the anatomical method of source placement, and the duration the implant remains inside the patient [1].

3.2.1 Classification by Dose Rate

- **Low Dose Rate (LDR):** Operates within a dose rate of 0.4 to 2 Gy/h. Radiation is administered continuously over an extended period (hours to several days), typically utilizing isotopes like ^{125}I and ^{137}Cs . It is a cornerstone for definitive prostate and traditional gynecological cancer treatments.
- **Medium Dose Rate (MDR):** Encompasses a transitional range of 2 to 12 Gy/h. Once utilized to emulate LDR biological effects over slightly shorter durations, it has largely been superseded in modern clinics by HDR techniques.
- **High Dose Rate (HDR):** Defined by a delivery rate exceeding 12 Gy/h (often much higher in practice). HDR utilizes sophisticated remote afterloading systems to shuttle a high-activity source (e.g., ^{192}Ir , ^{60}Co) into the patient for brief, fractionated sessions lasting mere minutes. It is the dominant modality for cervical, endobronchial, and breast brachytherapy.

3.2.2 Classification by Source Placement

- **Intracavitary:** Radioactive applicators are inserted into existing natural anatomical cavities, such as the uterine cavity, vaginal vault, or esophagus.
- **Interstitial:** Radioactive seeds or flexible source-guiding catheters are surgically driven directly into the solid tumor matrix. This invasive technique is routinely applied to prostate, breast, and soft tissue sarcomas.
- **Surface (Mould/Plaque):** Custom-fabricated surface applicators or moulds hold the radioactive material directly against the external epidermis, primarily targeting superficial cutaneous malignancies or ocular melanomas.
- **Intraluminal:** The source delivery system is threaded through a lumen, such as a bronchus or biliary duct, to treat intraluminal obstructions.

3.2.3 Classification by Implant Duration

- **Temporary Implants:** The radioactive payload is positioned for a predetermined timeframe and subsequently retracted. This encompasses all HDR and afterloaded LDR modalities.
- **Permanent Implants:** Tiny, biologically inert radioactive "seeds" (e.g., ^{125}I or ^{103}Pd) are permanently deposited within the target tissue. They continuously deliver their total integrated dose as they undergo natural radioactive decay.

3.3 Source Specification Methods Currently in Use

The specification of brachytherapy source strength has historically evolved through several distinct methods, reflecting advancements in dosimetric understanding and the need for standardized clinical communication. As detailed in AAPM Report No. 21 [5], the primary methods currently or historically utilized encompass the activity of the radionuclide, exposure rate at a reference point, equivalent mass of radium, and air-kerma rate.

3.3.1 Activity of the Radionuclide

The most fundamental, yet clinically challenging, specification is the absolute activity of the contained radionuclide. While it represents the true physical number of disintegrations per unit time (measured in Becquerels or Curies), specifying a source solely by its contained activity is often inadequate for clinical dosimetry. The primary limitation is that identical contained activities of the same radionuclide can yield significantly different dose rates in tissue if they are encapsulated differently (e.g., varying wall thickness or material). Furthermore, self-absorption within the source material itself and filtration by the encapsulating wall are not accounted for when only contained activity is specified [5]. Therefore, clinical application requires knowledge of the "apparent activity," which is the activity of a hypothetical unencapsulated point source that would produce the same exposure rate in free space at the same distance.

3.3.2 Exposure Rate at a Reference Point in Free Space

To circumvent the ambiguities of contained activity and encapsulation filtration, specifying the source by the exposure rate it produces at a specific distance in free space became a preferred method. This approach directly relates the source specification to the radiation field it generates, which is the quantity of clinical interest. The exposure rate, typically expressed in Roentgens per hour (R/h) or $C \cdot kg^{-1} \cdot h^{-1}$, is usually specified at a reference distance of 1 meter. This method intrinsically accounts for self-absorption and capsule attenuation, providing a more robust basis for dosimetric calculations [5].

3.3.3 Equivalent Mass and Apparent Activity

Historically, before the widespread adoption of modern dosimetric quantities, sources (particularly substitutes for Radium-226) were often specified by their "equivalent mass of radium." This is the mass of a point source of Radium-226 (encapsulated in 0.5 mm of platinum) that produces the same exposure rate at a given distance as the source in question. This metric allowed clinicians to utilize established radium dosimetry tables (like the Paterson-Parker system) for newer isotopes. However, as the use of diverse isotopes with varying photon spectra increased, this comparison became less rigorous and more

prone to error, especially when determining dose distributions in tissue near the source. Consequently, it has been largely deprecated in favor of more direct quantities [5].

3.3.4 Air-Kerma Rate at a Reference Point in Free Space

The current standard and most rigorous method recommended by modern protocols (including AAPM TG-32 and subsequently TG-43) is the specification by Air-Kerma Rate at a reference point in free space, or more formally, Air-Kerma Strength (S_K). This metric, fundamentally related to exposure but expressed in SI units (Gy/h or $\mu\text{Gy m}^2 \text{h}^{-1}$), represents the absolute air-kerma rate in vacuo due to photons of energy greater than a cutoff value (typically $\delta = 5 \text{ keV}$), at a distance d (usually 1 meter), multiplied by the square of that distance. This approach eliminates the historical reliance on exposure and radium equivalents, providing a direct, physical, and highly reproducible measure of the source's radiation output that seamlessly integrates into modern, model-based dose calculation algorithms [5].

3.4 Brachytherapy Radioactive Sources

3.4.1 Encapsulation and Design Integrity

To guarantee radiological safety and preclude the systemic dissemination of toxic radioisotopes into the patient's bloodstream, all clinical brachytherapy sources are strictly double-encapsulated. The enclosing capsules are typically micro-welded from robust, high-Z metals such as stainless steel, titanium (Ti), or platinum/iridium alloys. These specific materials are favored due to their superior mechanical resilience, minimal attenuation of therapeutic photons, and absolute chemical inertness (biocompatibility) [2].

3.4.2 Photonic Sources

The vast majority of brachytherapy procedures rely on photon-emitting radionuclides. These sources produce photon spectra with average energies generally spanning from 20 keV to 1.25 MeV. They are further divided into Low Energy (LE, $E_{avg} \leq 50 \text{ keV}$, e.g., ^{125}I) which are easily shielded and used for permanent implants, and High Energy (HE, $E_{avg} > 50 \text{ keV}$, e.g., ^{192}Ir , ^{60}Co) which demand substantial concrete or lead shielding and are standard for temporary HDR afterloading [1].

3.4.3 Source Strength Metrics

Traditional definitions of source strength relied on the "Radium-Equivalent Mass," comparing a given source's exposure rate to a historically standardized 1 mg ^{226}Ra source [6]. Modern TG-43 protocols, however, mandate the use of kerma-based quantities that directly relate to energy fluence [3].

Air-Kerma Strength (S_K): The fundamental contemporary metric. It represents the absolute air-kerma rate ($\dot{K}_\delta$) generated by the source in a scatter-free vacuum geometry, standardized to a specific distance d (usually 1 meter):

$$S_K = \dot{K}_\delta(d) \cdot d^2 \quad (1)$$

S_K is expressed in units of $\mu\text{Gy m}^2 \text{h}^{-1}$ (often denoted by the symbol U). The energy threshold δ (typically 5 keV) is applied to mathematically exclude low-energy contaminant photons (like Titanium characteristic X-rays from the capsule) that inflate the measured air-kerma without contributing any therapeutically useful dose at depths greater than a millimeter in tissue [4].

3.5 Brachytherapy Equipment: Siemens Healthineers BRAVOS

Modern High-Dose-Rate (HDR) brachytherapy delivery relies heavily on automated remote afterloader systems designed for maximum precision, safety, and operational efficiency. The Siemens Healthineers BRAVOS™ system [7] exemplifies the current state-of-the-art in this domain. This system typically utilizes a high-activity ^{192}Ir source, securely stored within a heavily shielded internal safe when not in use.

The BRAVOS afterloader is characterized by its integration of advanced safety and positioning technologies. A primary feature is its sophisticated pre-treatment verification process. Before the radioactive source is deployed into the patient, a non-radioactive "dummy" check cable is driven through the transfer tubes and into the applicator to confirm that the entire path is clear of obstructions and that the applicator geometry matches the treatment plan. This critical step ensures that the active source will reach its intended dwell positions without incident. The system employs high-precision stepper motors and

encoder feedback loops to position the source with sub-millimeter accuracy at each programmed dwell location. Furthermore, the BRAVOS system integrates seamlessly with treatment planning software, allowing for complex, multi-catheter implants and highly conformal 3D dose distributions. The unit also includes redundant safety mechanisms, such as continuous internal radiation monitoring, automatic source retraction upon power failure or system error, and manual retraction capabilities, ensuring patient and staff safety throughout the procedure.

3.6 Dosimetry via Well-Type Ionization Chamber

The internationally recognized gold standard for the clinical verification of brachytherapy source strength is the re-entrant or well-type ionization chamber. The standard dosimetric assembly includes the vented well-chamber, a specialized source-positioning insert (holder), a highly sensitive precision electrometer, and triaxial cabling [1].

3.6.1 The Well Chamber Design

Constructed specifically to intercept sources in a near 4π geometry, the chamber features a cylindrical outer housing and a concentric inner well into which the source is inserted. It requires an unsealed, vented design to freely adapt to ambient atmospheric changes (temperature and pressure). The device incorporates a central collecting electrode, an outer polarizing electrode, and a guard electrode strictly configured to minimize leakage currents.

A critical functional characteristic of a well-chamber is its "sweet spot"—a localized axial region along the well's depth that yields the maximum and most uniform ionization response. The longitudinal extent of this sweet spot must comfortably exceed the physical active length of the source to guarantee measurement stability despite minor mechanical positioning uncertainties [2].

3.6.2 Measurement Formalism

When utilizing a calibrated well-chamber, the Reference Air Kerma Rate ($\dot{K}_R$ or RAKR) is derived directly from the measured electrometer ionization current:

$$\text{RAKR} = \dot{M}_{corr} \cdot N_K \quad (2)$$

Where $\dot{M}_{corr}$ is the raw electrometer current reading mathematically corrected for environmental and operational influence quantities (temperature, pressure, ion recombination, and polarity effects), and N_K represents the specific calibration coefficient (in $\text{Gy h}^{-1} \text{A}^{-1}$) provided by an accredited Secondary Standard Dosimetry Laboratory (SSDL) for that specific well-chamber, electrometer, and source isotope combination.

3.7 HDR Afterloader Mechanics

The HDR remote afterloader is an advanced, computer-driven electromechanical apparatus designed to securely store and precisely deploy a high-activity source (typically a 10-15 Ci ^{192}Ir source). Employing "stepping source" mechanics, a solitary miniaturized source is laser-welded to the terminal end of a flexible steel drive cable. The system sequentially advances the source through transfer tubes into the patient applicator, pausing at programmed "dwell positions" for specific "dwell times" to carve out highly customized, 3D conformal dose distributions [1].

The unit houses the source in a dense tungsten safe when idle and incorporates extensive redundant safety mechanisms. These include a "dummy" check cable (run prior to the active source to verify unobstructed transit), active internal radiation monitors, and multiple manual retraction backups (e.g., a physical hand crank) to rescue the source in the event of an electrical failure or mechanical jam [6].

4 Observation

Prior to taking primary measurements, the source positioning was validated using an external Cam-Scale verification system.

4.1 Determination of Maximum Response Position (Sweet Spot)

The well-type ionization chamber was integrated with the precision electrometer, with the polarity adjusted to match the specifications detailed in the chamber's calibration certificate. A calibrated thermometer and barometer were deployed nearby to log the prevailing ambient temperature and atmospheric pressure.

Following the connection of the HDR source channel to the afterloader system, the ^{192}Ir source was systematically stepped through various dwell positions within the chamber's internal well. The steady-state ionization current was recorded at each specific position to map the chamber's spatial response profile. This procedure identifies the 'sweet spot'—the precise longitudinal position yielding the absolute maximum and most stable dosimetric signal.

Table 1: Ionization current measured across varied dwell positions.

S.No	Dwell Pos. (cm)	Current (nA)	S.No	Dwell Pos. (cm)	Current (nA)
1	131.9	44.033	12	126.4	51.22
2	131.4	45.630	13	125.9	50.79
3	130.9	47.080	14	125.4	50.16
4	130.4	48.340	15	124.9	49.33
5	129.9	49.370	16	124.4	48.25
6	129.4	50.190	17	123.9	46.91
7	128.9	50.815	18	123.4	45.25
8	128.4	51.242	19	122.9	43.26
9	127.9	51.490	20	122.4	40.91
10	127.4	51.560	21	121.9	38.24
11	126.9	51.480			

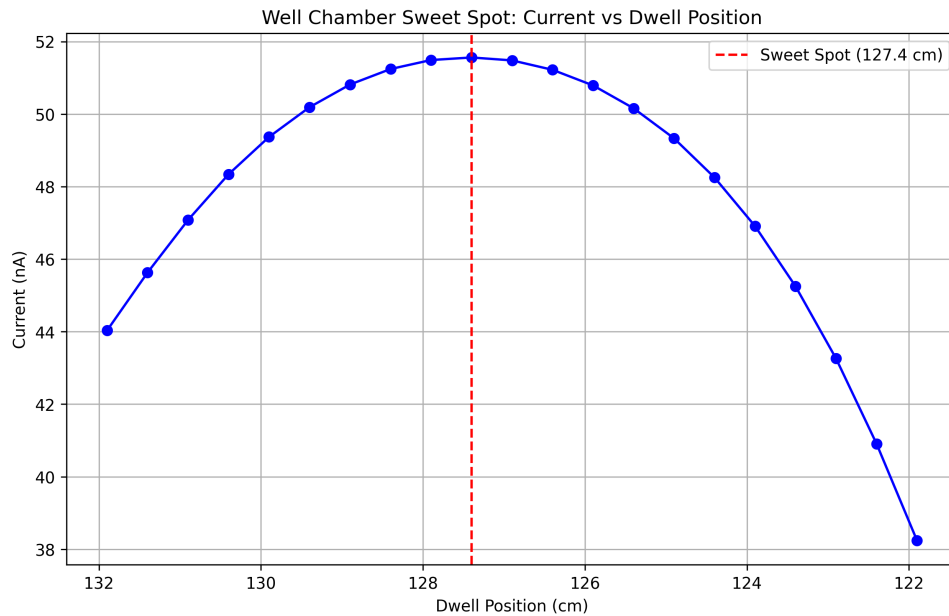


Figure 1: Response curve defining the sweet spot.

Analysis of the response curve clearly indicates a maximum signal intensity at a dwell position of 127.4 cm, generating a peak current of 51.56 nA. This location, mapped to dwell channel number 10, was subsequently utilized for all primary strength measurements.

4.2 Evaluation of Source Strength

With the source localized at the predetermined sweet spot (127.4 cm, channel 10), three independent readings of 20 seconds duration were recorded to ensure statistical reliability.

Table 2: Electrometer readings at the optimal sweet spot.

Dwell Pos.	Polarity (V)	Current (nA)			Avg. Current (nA)
		Trial 1	Trial 2	Trial 3	
10	-300	51.52	51.52	51.52	51.52
10	+150	51.44	51.44	51.44	51.44
10	+300	51.52	51.52	51.52	51.52

4.2.1 Derivation of Correction Factors

To derive the true meter reading, several necessary correction factors must be computed based on operational parameters.

1. Temperature and Pressure Correction (K_{TP}) Given local conditions of $T = 23.5^\circ\text{C}$ and $P = 1010$ mb against reference states $T_0 = 20^\circ\text{C}$ and $P_0 = 1013.2$ mb:

$$K_{TP} = \left(\frac{23.5 + 273.15}{20 + 273.15} \right) \cdot \left(\frac{1013.2}{1010} \right) = 1.015 \quad (3)$$

2. Polarity Correction (K_P) Utilizing readings from opposing polarities ($\pm 300\text{V}$) referenced to the calibration polarity ($+300\text{V}$):

$$K_P = \frac{51.52 + 51.52}{2 \times 51.52} = 1.000 \quad (4)$$

3. Ion Recombination Correction (K_S) Derived via the two-voltage technique (300V vs 150V):

$$K_S = \frac{4}{3} - \frac{51.52}{3 \times 51.44} = 1.0005 \quad (5)$$

4. Electrometer Factor (K_E) Given that the chamber and electrometer were calibrated as a unified pair, $K_E = 1.000$.

The final corrected reading ($\dot{M}_{corr}$) is:

$$\dot{M}_{corr} = 51.52 \text{ nA} \times 1.015 \times 1.000 \times 1.0005 \times 1.000 = 52.32 \text{ nA} \quad (6)$$

4.2.2 RAKR and Source Activity Computation

Using the provided calibration factor $N_K = 4.615 \times 10^5 \text{ Gy h}^{-1} \text{ A}^{-1}$:

$$\text{RAKR} = 52.32 \times 10^{-9} \text{ A} \times 4.615 \times 10^5 \text{ Gy h}^{-1} \text{ A}^{-1} = 24.1456 \text{ mGy h}^{-1} \quad (7)$$

Converting this to apparent Activity (A) utilizing the specific air-kerma rate constant:

$$\begin{aligned} \Gamma_\delta &= 4.07114 \text{ mGy m}^2 \text{ h}^{-1} \text{ Ci}^{-1} \\ A &= \frac{24.1456 \times (1)^2}{4.07114} = 5.931 \text{ Ci} \end{aligned} \quad (8)$$

4.3 Timer Error Determination

The intrinsic error associated with source transit and timer delay was evaluated using an interruption methodology over a 60-second exposure. The cumulative charge was compared between a continuous 60s run (R_1) and an equivalent run deliberately halted midway at 30s (R_2).

Table 3: Charge accumulation for timer error test.

Mode	Trial 1 (μC)	Trial 2 (μC)	Average (μC)
Uninterrupted (R_1)	3.103	3.103	3.103
Interrupted (R_2)	3.146	3.146	3.146

The resulting Timer Error is given by:

$$\text{Error} = \frac{(R_2 - R_1) \times t}{2 \times R_1 - R_2} = \frac{(3.146 - 3.103) \times 60}{2(3.103) - 3.146} = 0.8431 \text{ s} \quad (9)$$

4.4 Timer Linearity and End Error Assessment

Timer proportionality was checked across varying set durations relative to a known constant current ($I = 0.05148 \mu\text{A}$, established via a 300s dwell delivering $12.356 \mu\text{C}$ over a measured 240s window).

Table 4: Linearity check via variable dwell times.

Set Time (s)	Average Charge (μC)	Measured Time (s) = Charge/Current
60	3.103	60.27
120	6.190	120.24
180	9.277	180.20
240	12.365	240.19
300	15.456	300.23

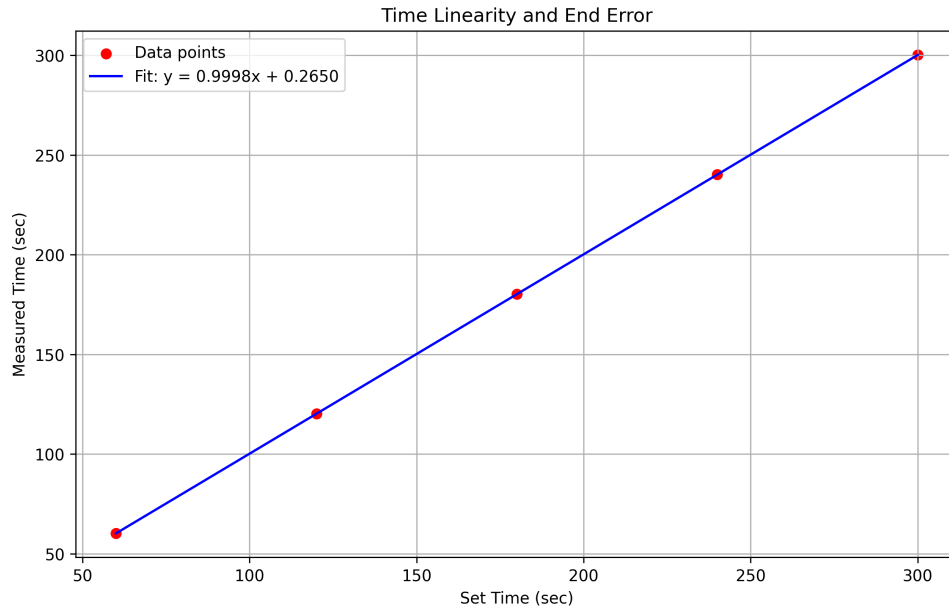


Figure 2: Timer linearity regression analysis.

Applying a linear fit to the measured data yields:

$$\text{Measured Time} = 0.9998 \cdot (\text{Set Time}) + 0.2650 \quad (10)$$

The derived metrics are therefore:

- **End Error (Y-intercept):** 0.2650 s
- **Percentage Time Linearity Error:** $|1 - 0.9998| \times 100\% = 0.02\%$

5 Conclusion

The quality assurance and dosimetric verification of the High Dose Rate (HDR) Brachytherapy unit was successfully completed using a well-type ionization chamber. The experimental procedure primarily aimed to determine the Reference Air Kerma Rate (RAKR) of the ^{192}Ir source, pinpoint the precise maximum response position (the "sweet spot") within the chamber, and systematically evaluate the electromechanical accuracy of the afterloader's treatment timer.

By meticulously stepping the source through the well-chamber, the sweet spot was experimentally identified at a dwell position of 127.4 cm (Channel 10), generating a stable peak ionization current of 51.56 nA. Measurements taken at this optimal position—after applying rigorous correction factors for ambient temperature, pressure, ion recombination, and polarity—yielded an absolute RAKR of $24.1456 \text{ mGy h}^{-1}$ at 1 meter. Using the specific gamma-ray constant for ^{192}Ir , this RAKR translates to an apparent source

activity of 5.931 Ci. This result is crucial for verifying that the current clinical output matches the manufacturer's decay-corrected specifications within strictly defined clinical tolerances (typically $\pm 3\%$).

In assessing the temporal fidelity of the HDR delivery system, the calculated intrinsic timer error was established at 0.8431 seconds. This minor discrepancy primarily accounts for the transit time required for the source to travel from the protective safe to the active dwell position and back. Furthermore, an extensive linearity check across multiple set dwell times spanning from 60 seconds to 300 seconds demonstrated an exceptionally high degree of proportionality. The calculated percentage time linearity error was a negligible 0.02%, coupled with an end error (y-intercept) of 0.2650 seconds.

In summary, the dosimetric calculations and temporal error analyses collectively confirm that both the physical strength of the ^{192}Ir source and the mechanical performance of the HDR afterloader's timing mechanism fall well within internationally accepted safety and quality assurance protocols. Consequently, the unit is deemed mechanically robust, dosimetrically accurate, and fundamentally safe for high-precision clinical brachytherapy treatments.

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